

2 Absciscic acid (ABA) is a plant hormone that influences the growth and development of plants.

Salicylic acid (SA) is a plant hormone that interacts with other plant hormones, such as ABA, to influence growth and development in plants.

Scientists investigated the effects of ABA and SA on shoot growth in seedlings of rice, *Oryza sativa*.

The scientists germinated some rice grains and divided the young seedlings into four batches, **A** to **D**, each containing 9 seedlings of the same length. The seedlings were put into trays containing a seedling growth medium. The shoots in each batch of seedlings were sprayed once with a different solution:

batch **A** – water

batch **B** – $2\mu\text{mol dm}^{-3}$ ABA

batch **C** – $2\mu\text{mol dm}^{-3}$ ABA + 1.0mmol dm^{-3} SA

batch **D** – 1.0mmol dm^{-3} SA

The seedlings were then put into a growth chamber with constant conditions of light, temperature and humidity. The length of the shoot on each seedling was measured one day after spraying and five days after spraying.

Fig. 2.1 shows two seedlings taken at random from each batch five days after being sprayed.

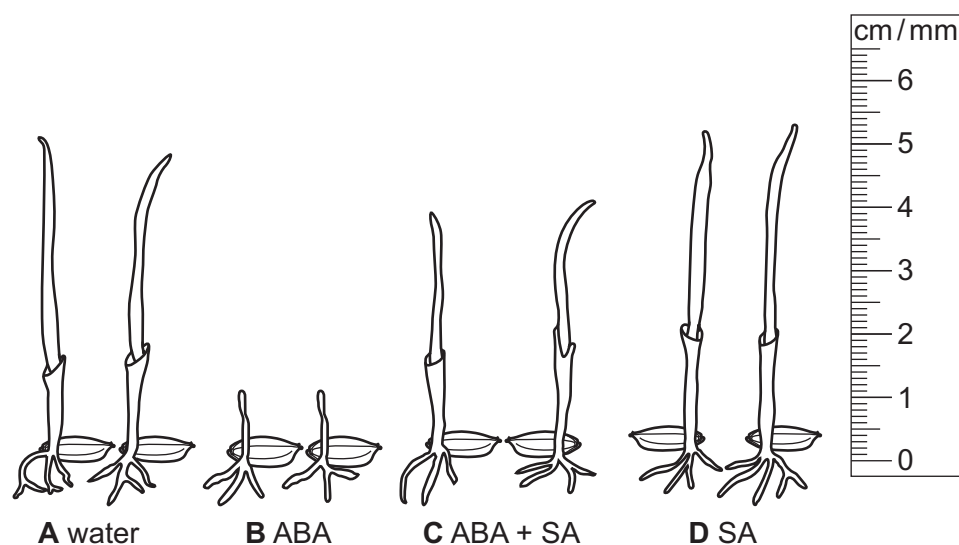


Fig. 2.1

The mean results of the investigation are shown in Fig. 2.2.

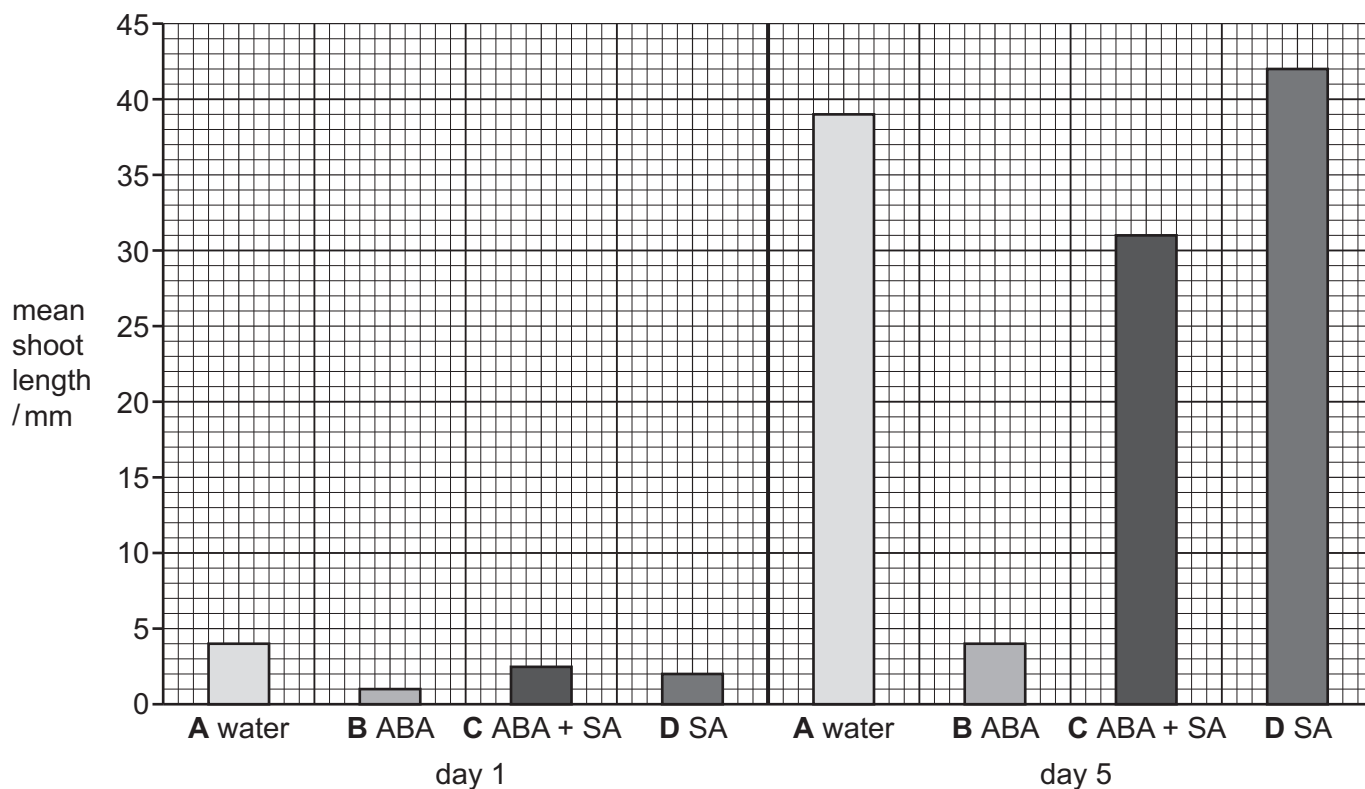


Fig. 2.2

(a) Seedlings do not always grow straight.

Suggest how the measurements of shoot length of the rice seedlings can be made as accurately as possible.

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- (b) There were nine seedlings in each batch.

The standard deviation for the length of seedlings in batch **D** (sprayed with SA only) at day 5 is 5 mm.

The formula for calculating standard error (SE) is:

$$SE = \frac{s}{\sqrt{n}} \quad \begin{array}{l} s = \text{sample standard deviation} \\ n = \text{sample size (number of observations)} \end{array}$$

The formula for calculating the 95% confidence interval (95% CI) is:

$$95\% \text{ CI} = \bar{x} \pm (2 \times SE) \quad \bar{x} = \text{mean}$$

- (i) Calculate the 95% CI for batch **D** (treated with SA only) at day 5.

95% CI = mm [1]

- (ii) The values for 95% CI were used to add error bars for batches **C** and **D** at day 5. These error bars did not overlap. State what can be concluded from this.

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- (c) Use the information in Fig. 2.2 to suggest **and** explain the effects of ABA and SA on the growth of the rice seedlings over five days.

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(d) The scientists delivered the plant hormones to the rice seedlings by spraying the shoots.

A student suggested that the rice seedlings could take up ABA and SA in the roots and transport the hormones to tissues in the shoot to give similar results to those in Fig. 2.2.

Outline a method to test this suggestion.

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Question 2 continues on page 12

- (e) ABA and SA are reported in the scientific literature to be involved in the control of the mitotic cell cycle.

Cyclin-dependent kinases (CDKs) are enzymes involved in the control of the mitotic cycle. The gene *CDKB2* codes for one of these kinase enzymes.

Genes *KRP4* and *CycD6* code for proteins that regulate the activity of CDKs.

The scientists investigated the effect of ABA and SA on gene expression.

Microarrays were used to measure the relative quantities of mRNA produced by transcription of the three genes, *CDKB2*, *KRP4* and *CycD6*, in the cells of the seedlings that were sprayed by ABA and SA.

The relative quantity of mRNA produced is a measure of the level of gene expression.

Fig. 2.3 shows the effects of ABA alone, SA alone and ABA and SA combined on the expression of the three genes.

Error bars show **standard deviation** ($n = 3$).

- (i) Complete Table 2.1 to show how the expression of the three genes is affected by the presence of ABA alone and SA alone, shown in Fig. 2.3.

Use **only** these descriptions to complete the table:

large increase small increase large decrease small decrease no change

Table 2.1

gene	effect on gene expression	
	ABA only	SA only
<i>CDKB2</i>		
<i>KRP4</i>		
<i>CycD6</i>		

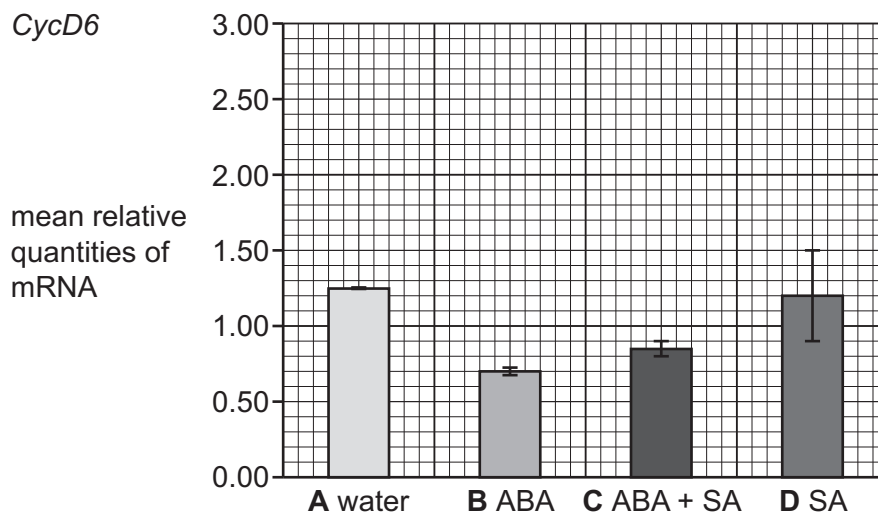
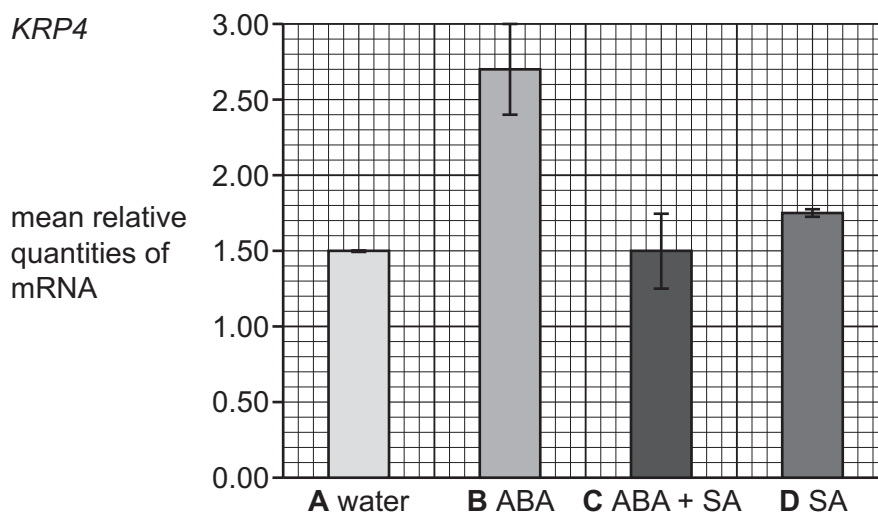
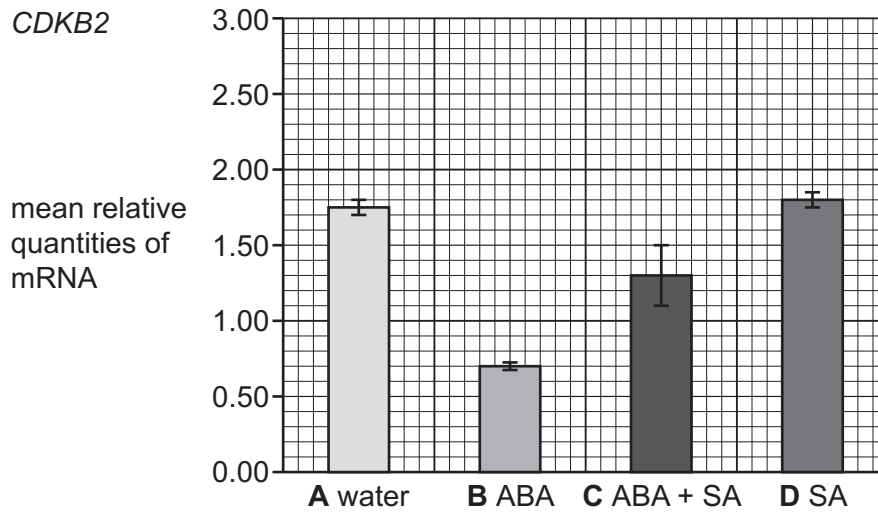


Fig. 2.3

- (ii) Fig. 2.3 also shows the results of spraying the seedlings with the mixture of ABA and SA (batch **C**) on the expression of the three genes.

The scientists made two conclusions about the effects of the mixture of ABA and SA (batch **C**) on the expression of the genes *KRP4* and *CycD6*.

Conclusion 1: SA counteracts the stimulation of gene expression of *KRP4* by ABA.

Conclusion 2: SA does not reverse the effect of ABA on gene expression of *CycD6*.

State **and** explain whether the evidence from Fig. 2.3 **fully supports**, **partially supports** or **does not support** each of these conclusions.

conclusion 1

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conclusion 2

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[2]

[Total: 14]